

US ECONOMICS ANALYST

The Economic Risks from Cyberattacks

- The rapidly progressing ability of AI models to identify and exploit vulnerabilities in the digital infrastructure that underpins the economy raises new concerns about the future of cyber risk. We provide an overview of cyberattacks and cybersecurity measures, outline how new AI models lower the barriers to successful cyberattacks but also improve cyber security capabilities, and assess the potential macroeconomic costs of AI-facilitated cyberattacks.
- Cyberattacks and their economic costs have increased substantially in recent years. Many of the most notable attacks on US companies involved data theft or deletion, unauthorized access of documents or machines, and ransomware, and the most extreme have imposed costs on targeted companies of up to \$1bn. Some of the reported increase in attacks reflects better cyber threat detection as companies have adopted stronger cybersecurity measures that alert them to a larger share of cyberattacks and help to address vulnerabilities before they can be exploited.
- Cybercriminals have already begun using AI for cyberattacks. In many cases, this involved criminals using AI models to access sensitive information, assist with espionage, or produce and distribute malware. But the latest models significantly raise the bar in terms of model intelligence and reasoning, software engineering capabilities, and vulnerability detection and exploitation capabilities, enabling them to execute much more complex and difficult to defend cyberattacks than prior models even without oversight from experienced human hackers.
- AI models can also improve cybersecurity in two ways. First, AI agents can help detect and respond to cyber threats faster, reducing the risk that malicious activities go unnoticed and progress into full-scale cyberattacks. Second, AI models can simulate cyberattacks by searching for and attempting to exploit vulnerabilities in existing infrastructure, allowing companies to address these vulnerabilities before they are exploited by cybercriminals. Early studies of the impact of AI on net cyber risk suggest that new AI models will likely benefit attackers more than defenders in the near term, which will likely result in faster, larger scale, and more sophisticated cyberattacks as new models are released to the public, but that the balance could favor defenders in the long term.
- How do cyberattacks affect the economy? Many cyberattacks result in modest direct financial loss—monetary losses from cyberattacks reported to the FBI averaged \$20k per attack in 2025, or \$40k for AI-related attacks, though some

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attacks on critical infrastructure have imposed costs of over \$1bn. But direct monetary losses are only a small part of the total cost of cyberattacks, which includes lost productivity, downstream effects on other businesses, regulatory fines, and cybersecurity spending. Accounting for these indirect costs, we estimate that the total cost of cyberattacks in the US was roughly \$300bn in 2025, or roughly 1% of GDP. It is too early to confidently estimate exactly how much AI-facilitated cyberattacks will boost these costs going forward, but more frequent cyberattacks, greater spending on AI-related cybersecurity, and increased regulatory costs will likely boost costs significantly in the coming years.

The Economic Risks from Cyberattacks

The rapidly progressing ability of AI models to identify and exploit vulnerabilities in the digital infrastructure that underpins the economy raises new concerns about the future of cyber risk. In this *Analyst*, we provide an overview of cyberattacks and cybersecurity measures, outline how new AI models lower the barriers to successful cyberattacks but also improve cyber security capabilities, and assess the potential macroeconomic costs of AI-facilitated cyberattacks.

Understanding Cyber Risk

Q: What are cyberattacks?

A: Cyberattacks are malicious attempts to harm an entity via its information and communications technologies.

Data from Google's Threat Intelligence Group suggests that attacks against zero-day vulnerabilities—those that were previously unknown to developers, meaning developers have had zero days to address the vulnerability—have become significantly more common over the past 5 years and have increasingly targeted technologies such as cloud services or other software systems used by large companies to run operations.¹

Exhibit 1: A Taxonomy of Common Cyberattacks

Taxonomy of Common Cyberattacks	
Type	Description
Denial-of-Service (DoS)	Flood servers with traffic to exhaust bandwidth or consume finite resources.
Malware	Software designed to cause damage to IT devices or steal data. Ransomware—a type of malware—denies users access to data or systems, after which the actor demands the payment of a ransom in exchange for the user regaining access.
Injection	An attack where a vulnerability in an application allows attackers to inject code into a program or query in order to execute remote commands.
Phishing	The practice of stealing sensitive data by sending fraudulent emails, texts, or audio that appear to be from a trustworthy source.
Brute Force	Trial and error method of guessing credentials to gain access to systems or data.
Zero-Day Attacks	An attack against a software or hardware vulnerability that has not been previously discovered.
N-Day Attacks	An attack against a software or hardware vulnerability that has been discovered and addressed by some but not all relevant parties.

Source: Goldman Sachs Global Investment Research

Q: What are some examples of recent cyberattacks?

A: Many of the most notable cyberattacks in recent years involved data theft or deletion, unauthorized access of documents or machines, or ransomware.

Even attacks with explicit financial motives, such as ransomware attacks, often also involve data theft. While it is often difficult to attribute cyberattacks to specific groups, postmortem analysis suggests that large attacks are often carried out by state actors working alongside criminal organizations.

¹ “Look What You Made Us Patch: 2025 Zero-Days in Review.” Google Threat Intelligence Group. March 5, 2026.

Exhibit 2: Many of the Most Notable Cyberattacks in Recent Years Involved Data Theft or Deletion, Unauthorized Access of Documents or Machines, and Ransomware

Recent High-Profile Cyber Attacks			
Date	Attacker	Victim	Description
Dec. 2020	Russia's Foreign Intelligence Service and a Russian hacker group	SolarWinds	Hackers hid malicious code in a file that was included with software updates, giving the hackers remote access to infected computers.
Mar. 2021	Chinese cyber espionage group, Hafnium	Microsoft Exchange Server	Hackers used several zero-day flaws to gain full access to emails and passwords, administrator privileges, and connected devices.
May 2021	Russian hacker group, DarkSide	Colonial Pipeline	Hackers used ransomware to access and lock computerized equipment managing the pipeline.
Sep. 2023	Hacker group, Scattered Spider	Caesars Entertainment and MGM Resorts	Hackers used a phishing campaign against an MGM employee and infiltrated a compromised IT support vendor at Caesars and spread ransomware across both companies' networks.
Feb. 2024	Russian cybercrime group, BlackCat	Change Healthcare	Hackers used ransomware to block doctors from getting paid and patients from getting prescriptions.
Jul. 2025	Suspected to be a US- or UK-based hacker group	Allianz Life Insurance	Hackers accessed a third-party cloud system to acquire insurance customers' data.
Mar 2026	Iran-linked hacker group	Stryker	Hackers compromised an administrative account and remotely wiped a large number of devices.

Source: Goldman Sachs Global Investment Research, Department of Homeland Security, Government Accountability Office, NPR

Q: How frequent are cyberattacks?

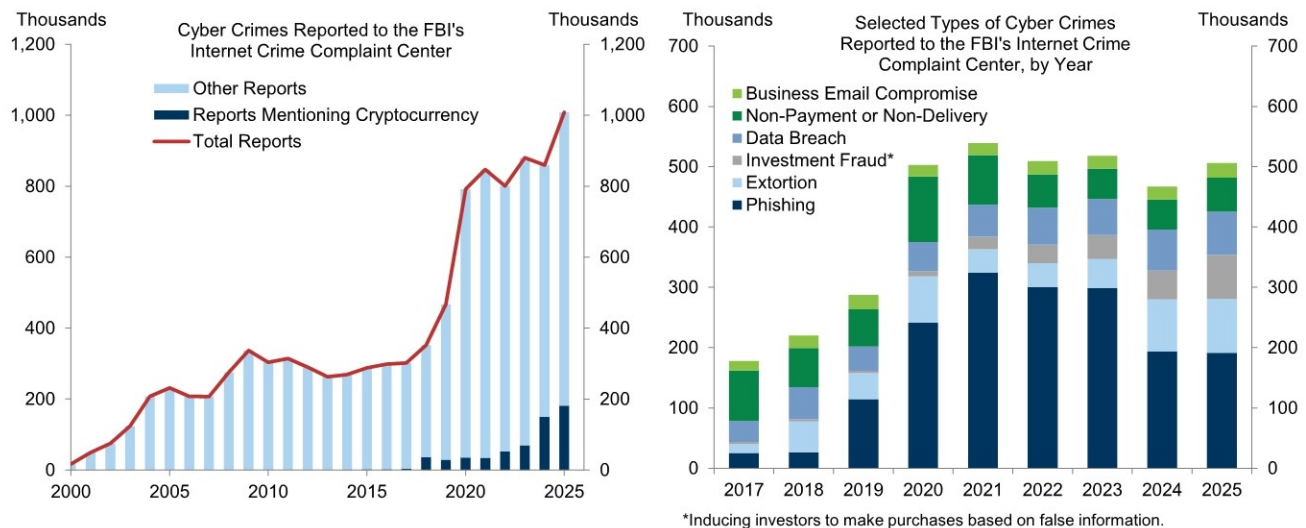
A: Over one million cyber incidents were reported to the FBI's Internet Crime Complaint Center (IC3) in 2025. However, this probably understates the prevalence because cyberattacks frequently go unnoticed and because there is little incentive for firms to report attacks.

Some of the reported increase in attacks reflects better cyber threat detection as companies have adopted stronger cybersecurity measures that allow them to address vulnerabilities before they can be exploited.

The sharp increase in reports in 2020 reflected the sudden increased dependence on technology during the pandemic lockdowns.² 40% of the increase in reports in 2020 reflected a more than doubling of reported phishing scams—one of the main types of attacks that targets end-users.

² 2020 Internet Crime Report. FBI Internet Crime Complaint Center. March 2021.

Exhibit 3: Cyber Crimes Reported to the FBI Have Tripled Since the 2010s, with Phishing Attacks Growing Particularly Sharply



Source: Goldman Sachs Global Investment Research, Department of Justice

Q: What cybersecurity measures exist and how are they being used to prevent cyberattacks?

A: Cybersecurity measures ranging from simple use of strong passwords to AI-driven security automation lower the cost, duration, and frequency of cyberattacks.

Annual cybersecurity disclosures now required by the SEC provide some insight into cybersecurity adoption among private-sector companies. Analysis of these disclosures published by Harvard indicates that companies most commonly implement access management, vulnerability scanning, threat intelligence, employee training, and security monitoring.³

Industry reports suggest that adoption of cybersecurity tools and spending on cybersecurity has surged in recent years. That said, a study by the World Economic Forum showed that 35% of small firms felt they had insufficient cybersecurity (vs. 5% in 2022) compared to just 7% of large firms, and Verizon's 2025 Data Breach Investigation report showed that small businesses had more data breaches and were exposed to malware more often than large firms.

³ Gosnell Handler et al. "Cybersecurity Disclosure Overview: A Survey of Form 10-K Cybersecurity Disclosures by S&P 100 Companies." January 9, 2025.

Exhibit 4: Cybersecurity Measures Range from Simple Use of Strong Passwords to Protect Accounts to AI-Driven Security Automation; Adoption of Cybersecurity Measures Reduces the Costs of Cyberattacks

Common Cybersecurity Measures		
Measure	Explanation	Efficacy
Password Protection	Using strong passwords to protect users' accounts and devices.	Focus on stronger passwords reduced the frequency of data breaches due to stolen credentials from 63% in 2016 to 22% in 2025.
Employee Training	Teaches employees how to detect phishing scams, create strong passwords, and handle data securely.	Reduced cost of data breach by \$190k on average.
Multi-Factor Authentication	Requires users to provide two or more verification factors to gain access to accounts.	Blocks 99% of bulk identity-based attacks.
Software Updates and Patching	Allows software developers to address known security vulnerabilities before they are exploited.	Government agencies that patched severe vulnerabilities saved an average of \$4.5mn in remediation costs.
Firewalls	A system that monitors traffic on a network to prevent unauthorized access.	Reduced cost of data breach by \$160k on average.
Antivirus and Antimalware Software	Software that continuously scans devices to detect and block viruses and malware.	Reduced cost of data breach by \$160k on average.
Zero Trust Architecture	Requires all systems to be continuously authenticated and authorized for every access request.	Reduced cost of data breach by roughly \$1mn relative to organizations that did not use zero trust.
Identity and Access Management	Restricting access to data or systems to only authorized users.	Reduced cost of data breach by \$190k on average.
Endpoint Detection	Continuously monitors end-users' devices to detect and address cyber threats.	Reduced cost of data breach by \$170k on average.
Data Encryption	Converts data into an unreadable format such that only authorized users can access it.	Reduced cost of data breach by \$210k on average.
Incident Response Planning	Developing a plan to detect and address cyber threats when they arise.	Reduced cost of data breach by \$200k on average.
AI-Driven Security Automation	Machine learning algorithms automatically detect and respond to threats without human intervention.	Reduced cost of data breach by \$160k - \$225k on average. Reduced length of breach by 45-80 days.
Bug Bounty Programs	Companies pay ethical hackers to find vulnerabilities in their code or other infrastructure.	Thousands of companies have discovered previously unknown vulnerabilities via bug bounty programs.
Offensive Security Testing or AI-Simulated Attacks	A software developer or an AI model in a training environment is told to find vulnerabilities in code/infrastructure and attempt to exploit them to launch a full-scale cyber attack.	Reduced cost of data breach by \$180 - \$225k on average.

Source: Goldman Sachs Global Investment Research, IBM, Microsoft, Department of Homeland Security

Understanding AI-Related Cyber Risk

Q: How are AI-facilitated cyberattacks different from human-led cyberattacks?

A: AI models enable attackers to work significantly faster and at a lower cost, evade threat detection measures, and find and exploit software vulnerabilities that prior testing and human attackers had not previously discovered.

While human-led cyberattacks often take days to months to execute, industry analysis suggests that AI-facilitated attacks can be completed within minutes and have become significantly faster over the past 5 years.⁴

AI models can also reduce the cost of an attack. As our technology equity analysts recently showed, AI models can complete coding tasks at a significantly lower cost than software engineers.

AI models can be used execute attacks with advanced evasion capabilities that make them difficult for threat detection measures to identify. For example, cybercriminals

⁴ CrowdStrike's 2026 Global Threat Report notes that the average cybercrime breakout time declined to 29 minutes in 2025 from 98 minutes in 2021.

have used AI to develop malware that can rewrite its own code to avoid detection. AI can also correct some of the tell-tale signs of phishing attacks like spelling and grammatical mistakes. Microsoft reported that AI-generated phishing emails are four times more effective than standard attempts. Relatedly, AI has enabled attackers to develop realistic vishing campaigns (fraudulent phone calls that sound like a trusted contact) and deepfake videos, two capabilities that were essentially impossible prior to the advent of AI.

As we will discuss in more detail below, AI models have demonstrated the ability to find and exploit zero-day vulnerabilities that had survived thousands or millions of rounds of vulnerability testing and had not been discovered by human attackers. This enables inexperienced attackers to use AI to locate vulnerabilities.

Q: How are cybercriminals using AI in their attacks?

A: Cybercriminals have already begun using AI models to access sensitive information, assist with espionage, and produce and distribute malware.

In their 2026 Global Threat Report, CrowdStrike reported that AI-enabled cybercriminals increased attacks by 89% year-over-year in 2025.⁵ Google, OpenAI, and Anthropic provide reports on attempted malicious uses of their AI models that they detected and prevented. As Exhibit 5 shows, recent attempted attacks were carried out disproportionately by state actors attempting to gather information on US entities.

Exhibit 5: Cybercriminals Have Begun Using AI Models to Facilitate Data Theft, Espionage, and Malware Creation and Distribution, Though Companies Have Been Able to Monitor and Block Many Attempts

State Actors' Attempted Misuse of AI Models
Gemini Usage - Reported and Stopped by Google
Conducting reconnaissance on US defense experts and organizations.
Researching US military and intelligence personnel and US military operations.
Researching US IT service providers.
Researching malware development and evasion techniques.
Crafting phishing campaigns.
Automating the process of accessing and stealing data from compromised Gmail accounts.
Attempting to learn about Gemini's underlying infrastructure and systems.
Adding encryption functions to existing code.
ChatGPT Usage - Reported and Stopped by OpenAI
Forging documents impersonating US officials to intimidate critics.
Claude Usage - Reported and Stopped by Anthropic
Apply, interview, and get hired for remote IT roles at US companies.
Private Actors' Attempted Misuse of AI Models
Gemini Usage - Reported and Stopped by Google
Creating an executable code file for a denial-of-service tool.
Producing malware.
Coding a Chrome infostealer.
Claude Usage - Reported and Stopped by Anthropic
Developing, marketing, and distributing ransomware with advanced evasion capabilities.
Running a web service that resells stolen credit cards.

Source: Goldman Sachs Global Investment Research, Company data

Q: What can new AI models do that previous models could not do or could not do well?

⁵ CrowdStrike 2026 Global Threat Report

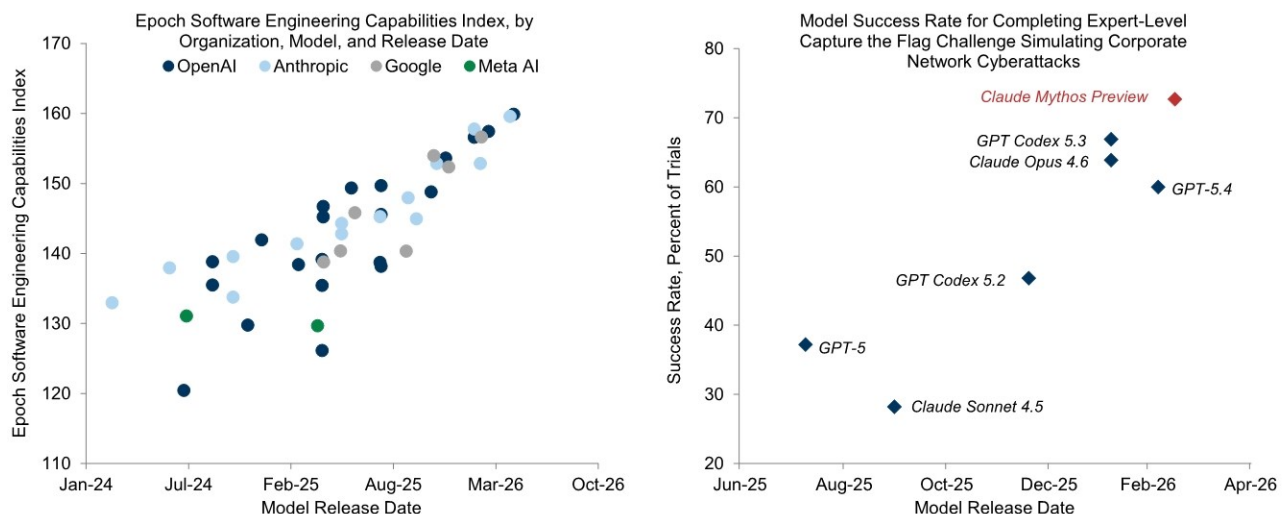
A: The latest models significantly raise the bar in terms of model intelligence and reasoning, software engineering capabilities, and vulnerability detection and exploitation capabilities, enabling them to execute much more complex and difficult-to-defend cyberattacks than prior models even without oversight from experienced human hackers.

AI models have become increasingly capable of completing software engineering tasks over time (Exhibit 6, left). Recently released models also score higher on tests of general model intelligence and reasoning and computer/hardware use.⁶

As an example, Anthropic's new Mythos model is notably better at finding and exploiting vulnerabilities in existing software than the company's prior models. In tests run by the UK government's AI Security Institute shown in the right panel of Exhibit 6, the Mythos model successfully executed a simulated cyberattack in over 70% of trials, a 5pp improvement over the model with the next highest success rate, OpenAI's GPT Codex 5.3. Of note, no model had successfully completed that specific test prior to April 2025.

As models' software engineering capabilities increase, so does their ability to assist cybercriminals with attacks or fully execute an attack with minimal human intervention. Anthropic noted that developers with no formal security training asked the Mythos model to find vulnerabilities and the model produced a complete, working exploit within hours, and that researchers have developed tools which allow the model to exploit vulnerabilities without any human intervention.

Exhibit 6: New AI Models Are Significantly Better at Software Engineering Tasks and Finding Vulnerabilities and Exploiting Them Than Prior Models



Source: Goldman Sachs Global Investment Research, Epoch AI, AI Security Institute

Understanding AI-Related Cyber Security

Q: How can AI models be used constructively to help companies improve their infrastructure and protect against future cyberattacks?

A: AI models can improve cybersecurity in two main ways: automatic threat detection and simulated cyberattacks.

⁶ Based on the Epoch Capabilities Index.

First, AI models can help detect cyber threats and help developers respond faster, reducing the risk that malicious activities go unnoticed and progress into full-scale cyberattacks. Data from IBM suggests that AI-automated cybersecurity measures reduced the cost of data breaches by \$1.9mn and that use of AI in cybersecurity significantly reduced the time it took organizations to identify and contain data breaches.

Second, AI models can simulate cyberattacks by searching for and attempting to exploit vulnerabilities in existing infrastructure, allowing companies to find and address these vulnerabilities before they are exploited by cybercriminals.

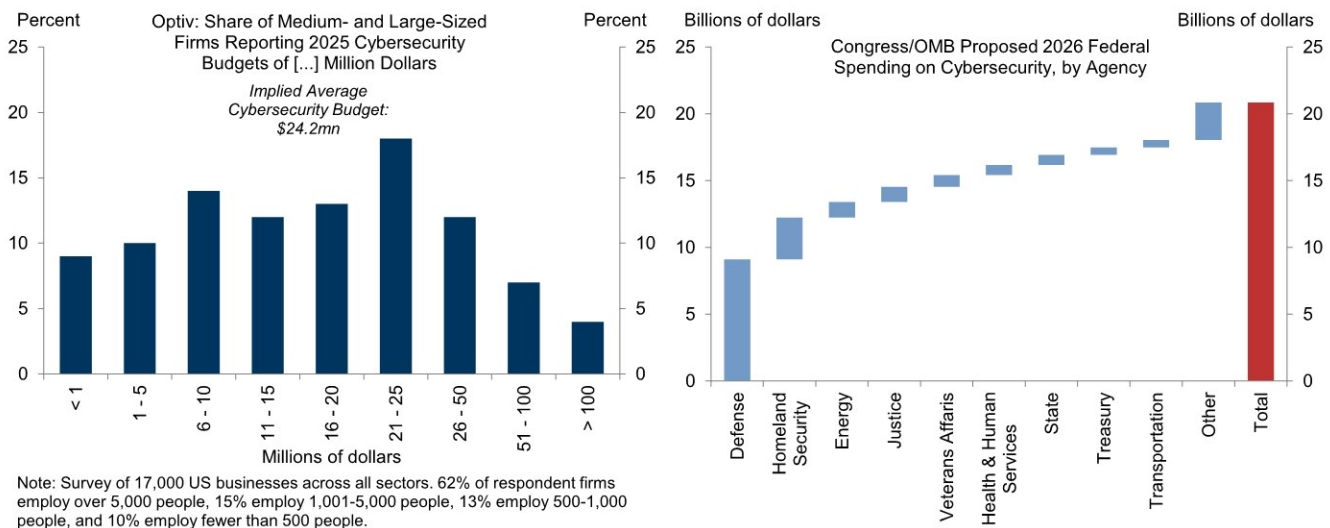
A study by researchers at Berkeley and UC Santa Barbara on the impact of AI on net cyber risk suggests that new AI models will likely benefit attackers more than defenders in the near term because many AI capabilities designed for cybersecurity can be repurposed for use by attackers, AI lowers the barriers for attackers, and AI lowers costs for attackers more than for defenders.⁷ While this will likely result in faster, larger scale, and more sophisticated cyberattacks in the near term, they argue that the balance could favor defenders in the long term as AI security measures reduce the number of exploitable flaws and raise the cost and difficulty for attackers to discover new vulnerabilities.

Understanding the Macroeconomic Impacts of Cyberattacks

Q: How much do organizations spend on cybersecurity?

A: Medium and large private sector companies reported nearly \$25mn in cybersecurity spending on average in 2025, and Congress and the Office of Management and Budget proposed roughly \$20bn in cybersecurity spending across agencies for 2026.

Exhibit 7: Medium- and Large-Sized Private Sector Companies Reported Roughly \$25mn in Cybersecurity Spending on Average in 2025; Congress and the Office of Management and Budget Proposed Roughly \$20bn in Cybersecurity Spending Across Federal Agencies for Fiscal Year 2026

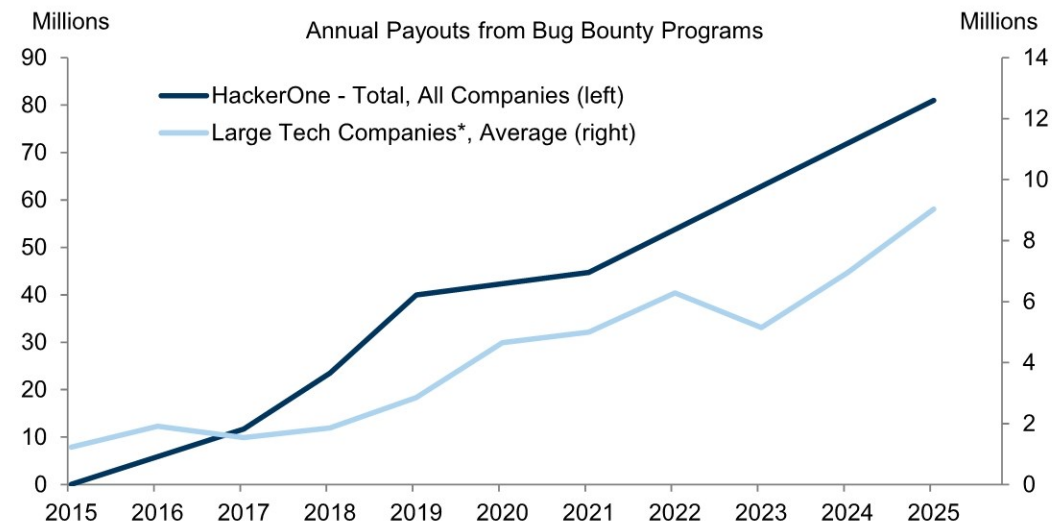


Source: Goldman Sachs Global Investment Research, Optiv, Office of Management and Budget, Congressional Research Service

⁷ "Frontier AI's Impact on the Cybersecurity Landscape." Potter et al. 2025.

Private sector businesses have also started hosting bug bounty campaigns where they pay ethical hackers to find vulnerabilities in their infrastructure. Over the past decade, companies have significantly increased incentives paid through these programs.

Exhibit 8: Companies Have Increased Incentives for Ethical Hackers to Find Bugs in Existing Infrastructure



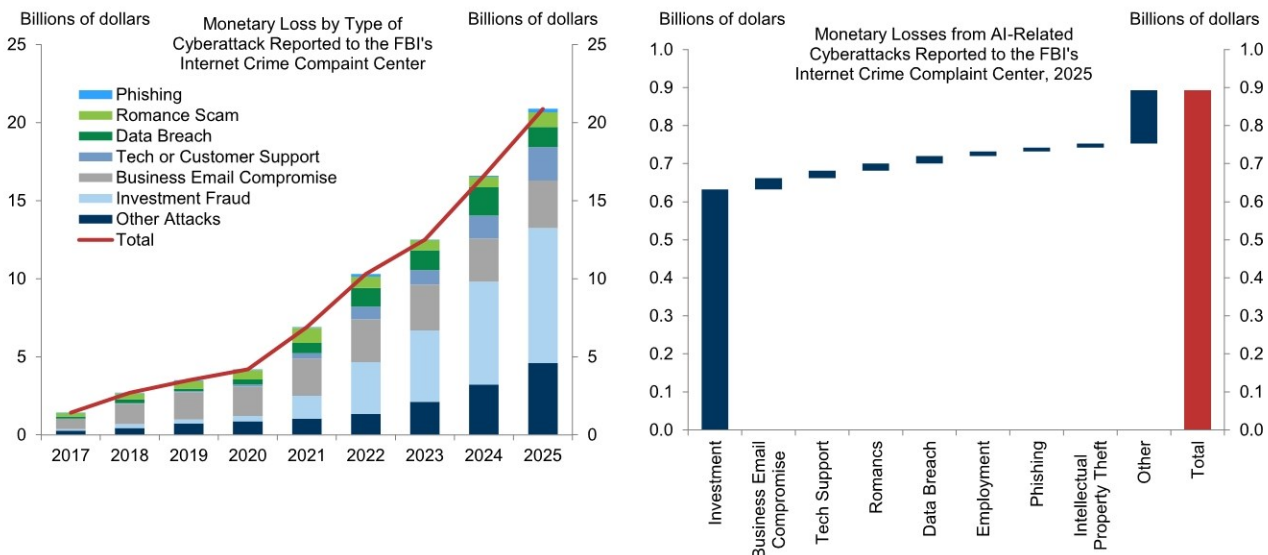
Source: Goldman Sachs Global Investment Research, HackerOne, Company data

Q: How costly are cyberattacks?

A: Many cyberattacks result in modest direct financial loss. But the direct monetary losses are only a small part of the total cost of cyberattacks, which includes lost productivity, downstream effects on other businesses, regulatory fines, cybersecurity spending, and intellectual property theft. Accounting for these indirect costs, we estimate that the total cost of cyberattacks in the US was roughly \$300bn in 2025.

Monetary losses from cybercrimes reported to the FBI have increased substantially over the past decade, averaging \$20k per attack in 2025, or \$40k for AI-related attacks. Investment fraud, in which an investor is induced to make purchases based on false information, accounted for 41% of total victim losses and 70% of losses from AI-related attacks in 2025. We view these reported losses as a floor for the costs of cyberattacks because many attacks are not reported at all and because many entities do not report any losses from ransomware attacks, creating an artificially low ransomware loss rate.

Exhibit 9: Direct Monetary Losses from Cyberattacks Have Increased Significantly Over the Past Decade, and Nearly a Billion in Losses from AI-Related Cyberattacks Were Already Reported to the FBI in 2025

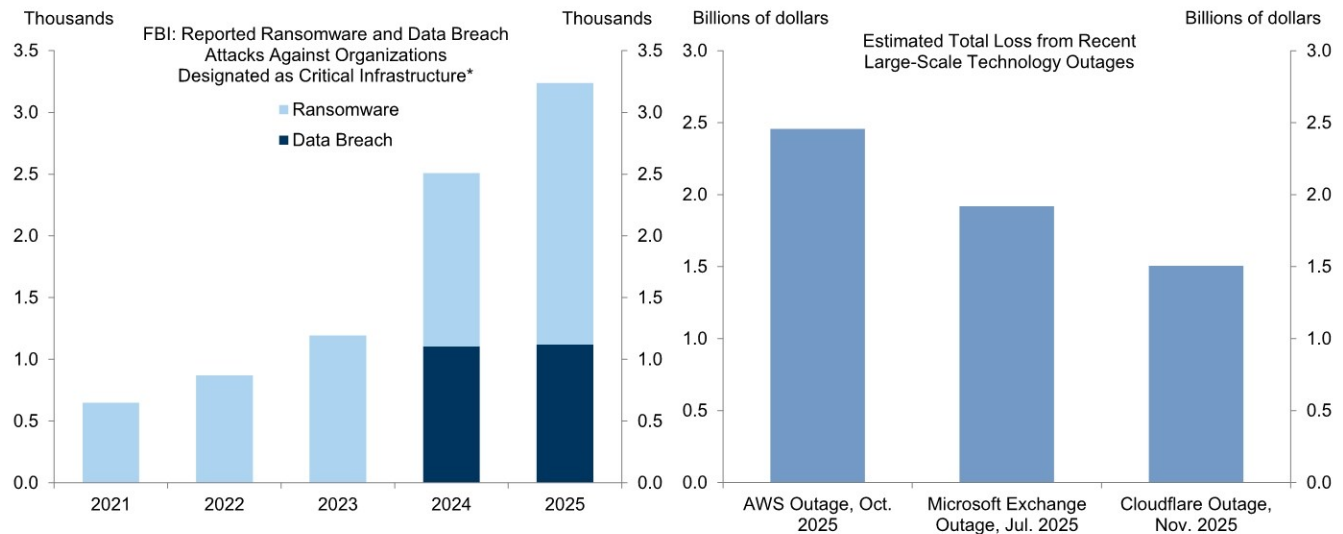


Source: Goldman Sachs Global Investment Research, Department of Justice

The FBI tracks ransomware attacks and data breaches against organizations designated as critical infrastructure whose assets, systems, and networks are considered so vital to the US that their incapacitation would have debilitating effects on national or economic security or public health and safety. Reported attacks against these sectors have become much more prevalent over the past 5 years (Exhibit 10, left).

While not caused by cyberattacks, recent large-scale technology outages give a sense of the potential financial impact of a cyberattack on a widely-used technology platform. Estimates suggest that total losses from each of these outages likely exceeded \$1bn. Costs from cyberattacks on these platforms would likely be substantially higher because recovery from an outage returns the platform to a known functional state, while a cyberattack requires extensive testing to determine functionality, loss of assets, data destruction, and prolonged trust-recovery costs.

Exhibit 10: Cyberattacks That Cause Outages in Sectors That Are Critical for US National or Economic Security Would Be Most Costly; While Not Caused by Cyberattacks, Recent Large-Scale Technology Outages Indicate the Potential Financial Impact of a Cyberattack on a Widely Used Technology Platform



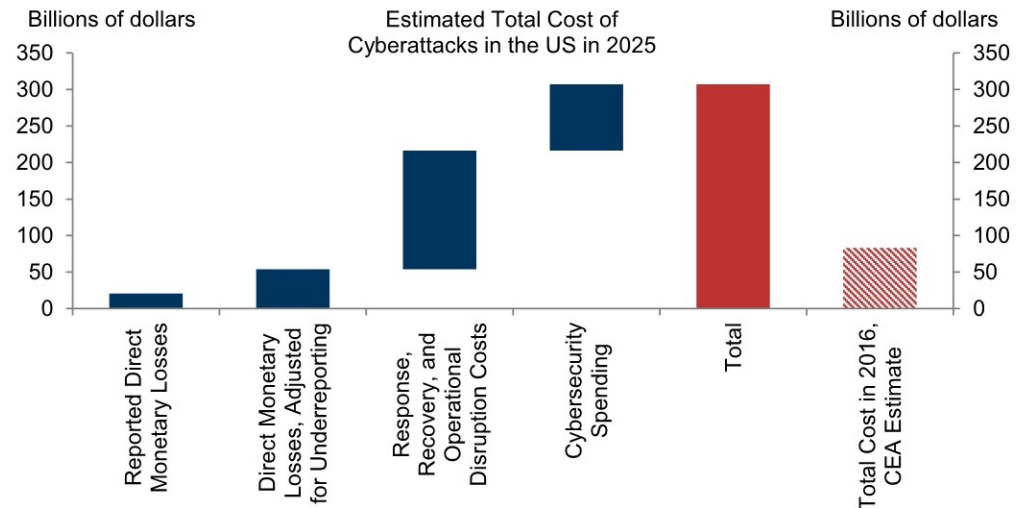
*According to the FBI, these are "sectors whose assets, systems, and networks, whether physical or virtual, are considered so vital to the US, their incapacitation or destruction would have a debilitating effect on national security, economic security, or public health and safety."

Source: Goldman Sachs Global Investment Research, Department of Justice

To estimate the total costs of cyberattacks in 2025, we first adjust the direct monetary losses reported to the FBI to account for underreporting. We then estimate various indirect costs including response and recovery, productivity loss, and cybersecurity spending as outlined in the Box below. Taken together, we estimate that the total cost of cyberattacks in the US was roughly \$300bn in 2025, significantly higher than the CEA's estimate of \$57–109bn in 2016. Cyberattacks also result in loss of intellectual property, the costs of which could be quite large but are difficult to estimate.

It is too early to confidently estimate exactly how much AI-facilitated cyberattacks will boost these costs going forward, but more frequent cyberattacks, greater spending on AI-related cybersecurity, and increased regulatory costs will boost costs significantly in the coming years.

Exhibit 11: Accounting for Underreporting of Attacks and Other Related Costs, We Estimate That the Annual Total Cost of Cyberattacks in the US Was Roughly \$300bn in 2025



Note: Excludes potential damage from theft of intellectual property.

Source: Goldman Sachs Global Investment Research

Jessica Rindels

Box: Calculations Used to Estimate Total Cost of Cyberattacks in the US in 2025

\$53bn in direct monetary losses adjusted for underreporting:

We adjust direct monetary losses reported to the FBI's Internet Crime Complaint Center in 2025 (\$20.9bn) for underreporting based on CEA analysis on the share of cyberattacks that go unreported or unnoticed. We add \$1.2bn to account for ransomware costs that are underreported in the FBI statistics.

\$163bn in response, recovery, and operational disruption costs:

We multiply IBM's 2025 average cost of a US data breach (\$10.2mn) by the sum of the number of data breaches in 2025 reported by the Identity Theft Center which measure data breaches involving individuals' personal data and one third of the 2025 data breaches in North America reported in Verizon's 2025 Data Breach Investigation report to capture breaches of company data in the US, both of which we adjust for unreported and unnoticed attacks. This totals \$109bn. We add \$44bn for downstream impacts on other businesses based on CEA and CSIS analysis on spillover effects of cyberattacks on other businesses and add \$10bn for lost productivity, based on our estimate that 5,000 employees across attacked and other impacted organizations experience diminished productivity for 4 hours per attack on average.

\$91bn in cybersecurity spending:

We take the sum of the \$20.8bn in federal spending on cybersecurity proposed by Congress and the Office of Management and Budget and \$70bn in private sector cybersecurity spending based on estimated global cybersecurity spending reported by Forrester Research.

The US Economic and Financial Outlook

THE US ECONOMIC AND FINANCIAL OUTLOOK

(% change on previous period, annualized, except where noted)

	2023	2024	2025	2026	2027	2025 Q4	Q1	2026 Q2	Q3	Q4	2027 Q1	Q2	Q3
OUTPUT AND SPENDING													
Real GDP	2.9	2.8	2.1	2.0	2.0	0.5	2.0	1.6	1.9	1.9	2.0	2.1	2.2
Real GDP (annual=Q4/Q4, quarterly=yoy)	3.4	2.4	2.0	1.9	2.2	2.0	2.7	2.1	1.5	1.9	1.9	2.0	2.1
Consumer Expenditures	2.6	2.9	2.6	1.8	1.8	1.9	1.6	0.9	1.4	1.5	1.9	2.0	2.1
Residential Fixed Investment	-7.8	3.2	-2.2	-3.1	1.9	-1.7	-8.0	1.0	1.5	1.5	2.2	2.2	2.2
Business Fixed Investment	7.3	2.9	4.1	5.9	4.9	2.4	10.4	6.0	5.2	4.6	4.8	4.7	4.7
Structures	16.7	1.1	-5.3	-2.9	3.7	-6.6	-6.7	2.2	4.0	4.0	4.0	3.5	3.5
Equipment	2.9	3.5	8.3	8.9	5.4	4.3	17.2	9.0	6.0	5.0	5.0	5.0	5.0
Intellectual Property Products	6.2	3.5	5.6	7.7	4.9	5.4	13.0	5.0	5.0	4.5	5.0	5.0	5.0
Federal Government	3.3	3.8	-1.2	-1.0	0.9	-16.7	9.4	0.0	1.0	1.0	1.0	1.0	1.0
State & Local Government	3.6	3.8	2.5	1.3	0.9	1.5	1.6	0.5	0.5	1.0	1.0	1.0	1.0
Net Exports (\$bn, '17)	-925	-1,033	-1,091	-1,096	-1,154	-969	-1,068	-1,096	-1,104	-1,116	-1,131	-1,146	-1,162
Inventory Investment (\$bn, '17)	47	44	29	25	49	-16	-8	28	35	45	45	48	50
Nominal GDP	6.7	5.3	5.0	5.9	4.6	4.2	5.6	7.6	4.8	4.4	4.4	4.3	4.3
Industrial Production, Mfg.	-1.0	-1.0	0.9	2.7	3.3	-3.1	5.7	4.4	3.5	3.2	3.1	3.2	3.3
HOUSING MARKET													
Housing Starts (units, thous)	1,421	1,371	1,356	1,286	1,322	1,323	1,280	1,290	1,280	1,292	1,304	1,316	1,328
New Home Sales (units, thous)	665	685	672	670	644	680	690	690	662	638	626	632	649
Existing Home Sales (units, thous)	4,103	4,067	4,076	4,212	4,375	4,157	4,169	4,189	4,223	4,267	4,310	4,353	4,396
Case-Shiller Home Prices (%yoy)*	5.3	3.8	0.6	0.8	2.2	0.6	0.3	0.8	0.7	0.8	1.1	1.4	1.8
INFLATION (% ch, yr/yr)													
Consumer Price Index (CPI)**	3.3	2.9	2.7	3.6	2.0	2.7	2.7	3.8	3.6	3.6	3.2	2.3	2.2
Core CPI **	3.9	3.2	2.6	2.1	2.0	2.7	2.5	2.7	2.3	2.2	2.0	1.8	1.8
Core PCE** †	3.1	3.0	3.0	2.7	2.0	2.9	3.1	3.2	3.1	2.9	2.3	2.0	1.9
LABOR MARKET													
Unemployment Rate (%)^	3.8	4.1	4.4	4.6	4.4	4.4	3.5	3.6	3.7	3.8	4.6	4.5	4.5
U6 Underemployment Rate (%)^	7.2	7.6	8.4	8.7	8.3	8.4	6.7	6.9	7.0	7.2	8.7	8.6	8.4
Payrolls (thous, monthly rate)	210	122	10	51	92	-39	500	348	408	253	70	97	100
Employment-Population Ratio (%)^	60.1	59.9	59.7	58.9	58.8	59.7	61.5	61.5	61.6	61.1	58.9	58.8	58.8
Labor Force Participation Rate (%)^	62.5	62.5	62.4	61.8	61.5	62.4	62.6	62.6	62.7	62.5	61.7	61.6	61.6
Average Hourly Earnings (%yoy)	4.2	5.4	4.6	4.0	3.9	4.4	4.2	4.1	3.9	3.8	3.9	4.0	4.0
GOVERNMENT FINANCE													
Federal Budget (FY, \$bn)	-1,694	-1,833	-1,775	-1,950	-2,050	--	--	--	--	--	--	--	--
FINANCIAL INDICATORS													
FF Target Range (Bottom-Top, %)^	5.25-5.5	4.25-4.5	3.5-3.75	3.25-3.5	3-3.25	3.5-3.75	3.5-3.75	3.5-3.75	3.5-3.75	3.25-3.5	3-3.25	3-3.25	3-3.25
10-Year Treasury Note^	3.88	4.58	4.18	4.10	4.15	4.18	4.30	4.30	4.20	4.10	4.10	4.10	4.15
Euro (€/€)^	1.11	1.04	1.17	1.18	1.20	1.17	1.15	1.16	1.19	1.18	1.19	1.20	1.20
Yen (\$/¥)^	141	157	157	158	141	157	159	158	158	158	156	141	101

* Weighted average of metro-level HPIs for 381 metro cities where the weights are dollar values of housing stock reported in the American Community Survey. Annual numbers are Q4/Q4.

** Annual inflation numbers are December year-on-year values. Quarterly values are Q4/Q4.

† PCE = Personal consumption expenditures. ^ Denotes end of period.

Note: Published figures in bold.

Source: Goldman Sachs Global Investment Research

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